

Case Study

Marketing Campaign Performance & Customer Response Analysis

Problem Statement

The marketing team has executed five customer marketing campaigns over the past year but lacks a clear understanding of which campaigns generated the strongest customer response and highest business value. While multiple campaigns have been launched, management cannot confidently identify which customer segments respond best, which purchase channels drive the most engagement, or whether marketing investments are being directed toward the right audience.

Without these insights, future campaign budgets and targeting strategies are largely based on assumptions rather than data, increasing the risk of lower campaign ROI and inefficient marketing spend.

The objective of this analysis is to evaluate campaign performance, understand customer purchasing behavior, identify the highest-value customer segments, and provide actionable recommendations to improve future marketing effectiveness.

Data Overview

- **Source:** Marketing Campaign Customer Dataset
- **Records:** ~2,240 Customers
- **Tools Used:** Excel, MySQL, Power BI

The dataset contains customer demographic information, income, education, family structure, product spending, purchase channels, website activity, and responses across five marketing campaigns.

Tool Stack

Excel (Data Cleaning) → MySQL (Business Analysis) → Power BI (Interactive Dashboard)

Business Questions

1. Which marketing campaign achieved the highest customer acceptance rate?
2. Which education segment responds best to marketing campaigns?
3. Which purchase channel generates the highest customer purchases?
4. Does higher income lead to higher customer spending and campaign acceptance?
5. Which product category contributes the highest revenue?
6. Do customers with children spend less and respond less to marketing campaigns?
7. Which Country generates the highest customer purchases?

Data Cleaning (Excel)

Performed data validation and quality checks before analysis.

Cleaning Steps

- Verified duplicate customer IDs.
- Checked for missing and blank values.
- Corrected inconsistent data formatting.
- Renamed columns for better readability.
- Verified numeric columns including Income and product spending.
- Standardized categorical values for Education and Marital Status.
- Created Total Children column (Kidhome + Teenhome).
- Created Total Spending column by combining all product categories.
- Verified campaign response fields.
- Imported cleaned dataset into MySQL for business analysis.

Analysis (SQL)

Key Findings

1. Which marketing campaign achieved the highest acceptance rate?

Campaign 4 recorded the highest acceptance rate of **7.46%**, while **Campaign 2** generated the lowest acceptance rate of **1.34%**. The significant variation suggests differences in campaign targeting, promotional offers, or customer relevance.

Campaign	Total_Acceptance	Acceptance_Rate_Pct
Campaign 4	167	7.46
Campaign 3	163	7.28
Campaign 1	144	6.43
Campaign 2	30	1.34

2. Which customer education segment responds best?

Customers with **PhD** demonstrated the highest campaign response rate at **6.75%** while also recording the highest average spending of **\$672.41**.

This indicates that higher education segments represent the most valuable audience for future marketing campaigns.

Education	total_customers	avg_income	avg_total_spend	total_campaign_responses	response_rate_pct
PhD	486	55567.69	672.41	164	6.75
Graduation	1127	52205.80	619.90	343	6.09
Master	370	52202.43	611.78	103	5.57
2n Cycle	203	46929.25	496.53	51	5.02
Basic	54	20306.26	81.80	6	2.22

3. Which purchase channel performs best?

Store purchases generated the highest total purchase volume with **12970 purchases**, demonstrating that physical stores remain the primary shopping channel.

However, **Catalog** purchases recorded the **highest average purchases per customer**, indicating stronger customer loyalty and higher-value transactions.

channel	total_purchases	avg_per_customer
Store	12970	5.79
Web	9150	4.08
Catalog	5963	2.66

4. Does higher income increase spending?

Customers earning more than **\$90,000** spent approximately **5 times** more than lower-income customers.

However, campaign acceptance rates did not increase proportionally with income, suggesting that customer engagement depends on factors beyond purchasing power alone.

income_band	total_customers	avg_income	avg_total_spend	campaign_response_rate_pct
1. Low (<30k)	394	20168.92	97.51	1.62
2. Mid (30k-60k)	1004	44508.40	298.24	3.09
3. High (60k-90k)	788	72387.20	1185.36	10.05
4. Very High (90k+)	54	113070.41	1575.41	31.11

5. Which product category generates the highest revenue?

Wine products generated total customer spending of **\$(X)**, making them the company's strongest-performing product category.

The large spending gap compared to other products highlights Wines as the primary driver of customer revenue.

product_category	total_spend	avg_spend_per_customer
Wines	680816	303.94
Meat Products	373968	166.95
Gold Products	98609	44.02
Fish Products	84057	37.53
Sweet Products	60621	27.06
Fruits	58917	26.30

6. Do customers with children behave differently?

Customers without children or teenagers at home spent approximately **2.5 times** more and recorded significantly higher campaign acceptance rates than households with two or more children.

Family size clearly influences both purchasing behavior and marketing responsiveness.

household_type	total_customers	avg_income	avg_total_spend	campaign_response_rate_pct	avg_web_visits_per_month
No Kids/Teens	638	65190.63	1106.03	12.01	3.48
1 Kid/Teen	1128	47247.97	472.73	3.92	5.94
2+ Kids/Teens	474	44077.15	249.15	2.66	6.32

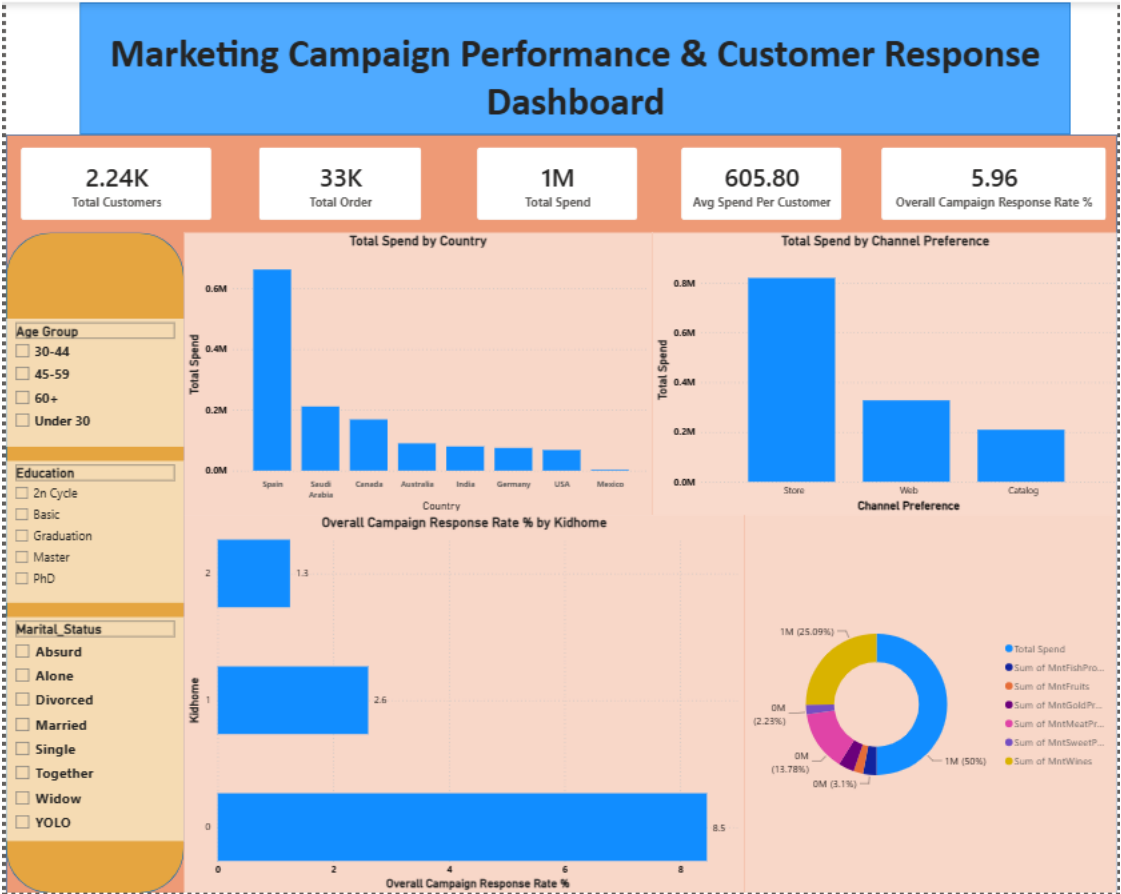
7. Which Country generates the highest customer purchases?

Spain generated the highest total spending with \$ **662220**, and having higher customers.

However, **Mexico** recorded the **highest Campaign Response Rate**, indicating stronger customer response.

Country	Total_Spending	total_customers	total_purchases	campaign_response_rate_pct
Spain	662220	1095	4382	6.48
Saudi Arabia	211071	337	1397	5.10
Canada	168532	268	1154	6.49
Australia	89763	160	654	4.38
India	79485	148	584	5.27
Germany	74913	120	477	6.33
USA	67882	109	484	4.77
Mexico	3122	3	18	6.67

Dashboard (Power BI)



Recommendations

1. Replicate the highest-performing campaign

Analysis the strategy behind **Campaign 4**, including its offer, timing, and target audience, and use it as the benchmark for future campaigns.

2. Priorities high-value customer segments

Increase marketing investment toward **Graduate and PhD** customer segments, as they generate both the highest spending and strongest campaign response.

3. Create dedicated Wine marketing campaigns

Wine products contribute the highest customer spending and should receive focused promotional campaigns, loyalty offers, and cross-selling opportunities.

4. Improve customer targeting

Allocate a greater proportion of campaign budgets toward households without children, as they demonstrate higher disposable income, greater spending, and stronger campaign engagement.

5. Expand Catalog marketing

Catalog customers show the highest average purchases per customer, indicating stronger loyalty and greater long-term customer value.

6. Personalize marketing campaigns

Rather than relying only on demographic characteristics such as income, use customer purchasing behavior and historical campaign responses to build more targeted and personalized marketing strategies.

7. Measure campaign success beyond acceptance rates

Track future campaigns using both customer response rates and post-campaign revenue to ensure marketing investments generate profitable business outcomes rather than simply increasing participation.